PredictiveGuard

Industrial Predictive Maintenance System

Model Training & Evaluation Report

Best Model: Logistic Regression

F1 Score: 0.1097

ROC AUC: 0.6832

Precision: 0.0599

Recall: 0.6503

Features Used: 18

Report Date: March 10, 2026

1. Executive Summary

This report presents the results of the predictive maintenance model training pipeline. A total of 4 models were trained and evaluated. The best performing model is Logistic Regression with an F1 score of 0.1097.

Best Model Metrics

Model	Logistic Regression
F1 Score	0.1097
ROC AUC	0.6832
Precision	0.0599
Recall	0.6503
Accuracy	0.6560
MCC	0.1138
Specificity	0.6562
Optimal Threshold	0.64
Features Used	18

2. Model Performance Comparison

The table below compares all trained models across key classification metrics. The best model is highlighted based on the primary optimization metric (F1 Score).

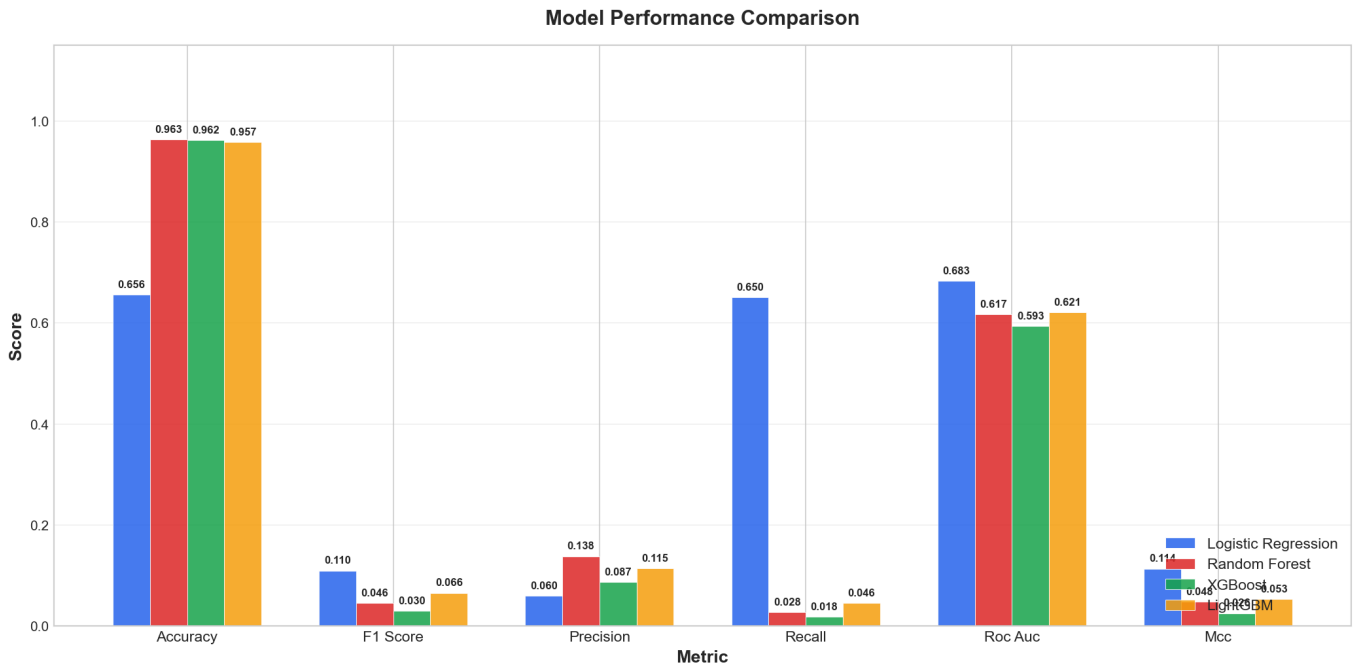
Model	F1	AUC	Precision	Recall	MCC	Time(s)
Logistic Regression *	0.1097	0.6832	0.0599	0.6503	0.1138	19.7
Random Forest	0.0460	0.6166	0.1385	0.0276	0.0482	1881.4
XGBoost	0.0304	0.5934	0.0870	0.0184	0.0255	158.7
LightGBM	0.0656	0.6213	0.1145	0.0460	0.0531	149.9

* Indicates the selected best model.

Cross-Validation Results

Model	CV F1 Mean	CV F1 Std	CV AUC Mean	CV AUC Std
Logistic Regression	0.5024	0.0057	0.6629	0.0040
Random Forest	0.9599	0.0019	0.9965	0.0003
XGBoost	0.9625	0.0029	0.9917	0.0008
LightGBM	0.9428	0.0028	0.9837	0.0007

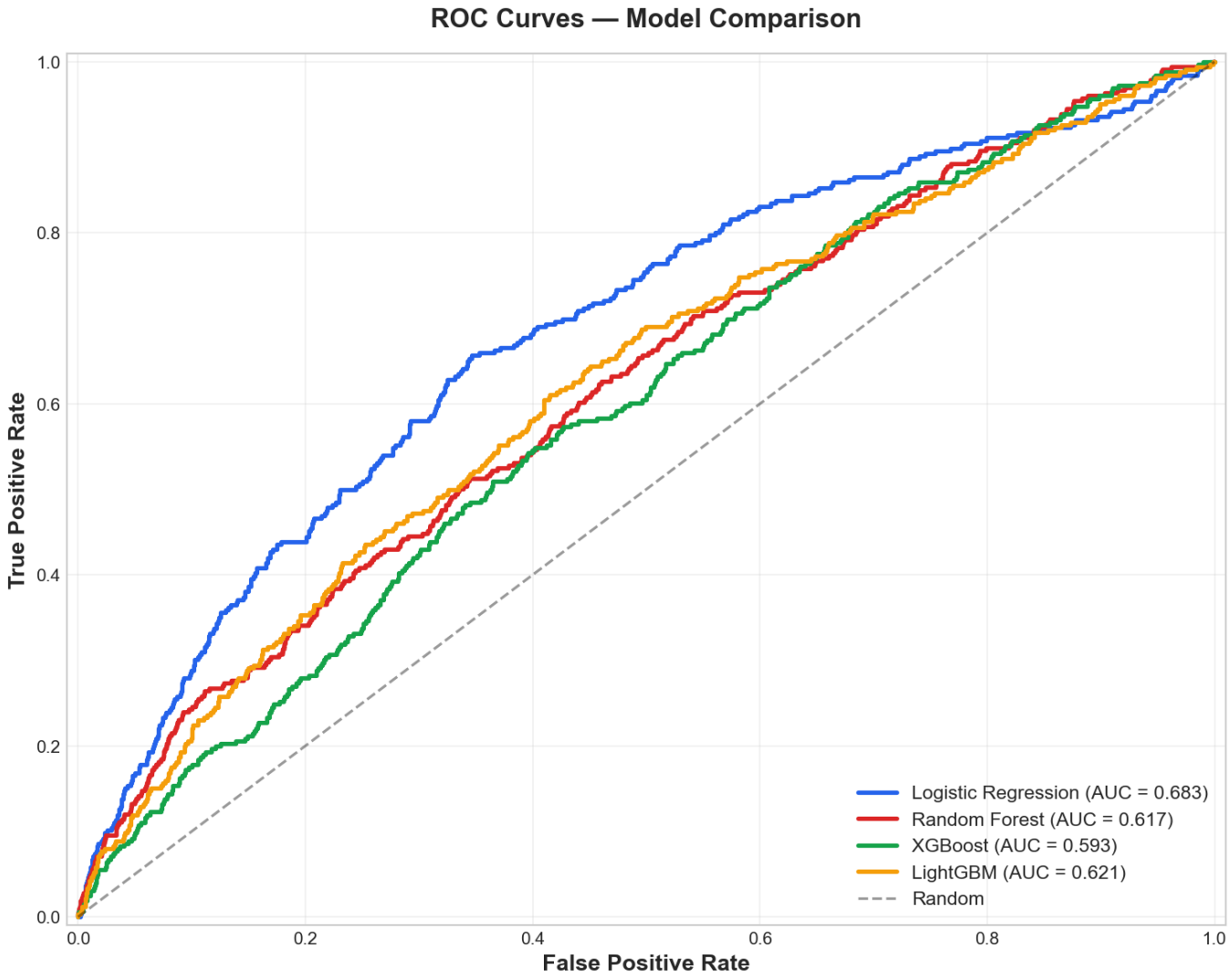
Visual Comparison



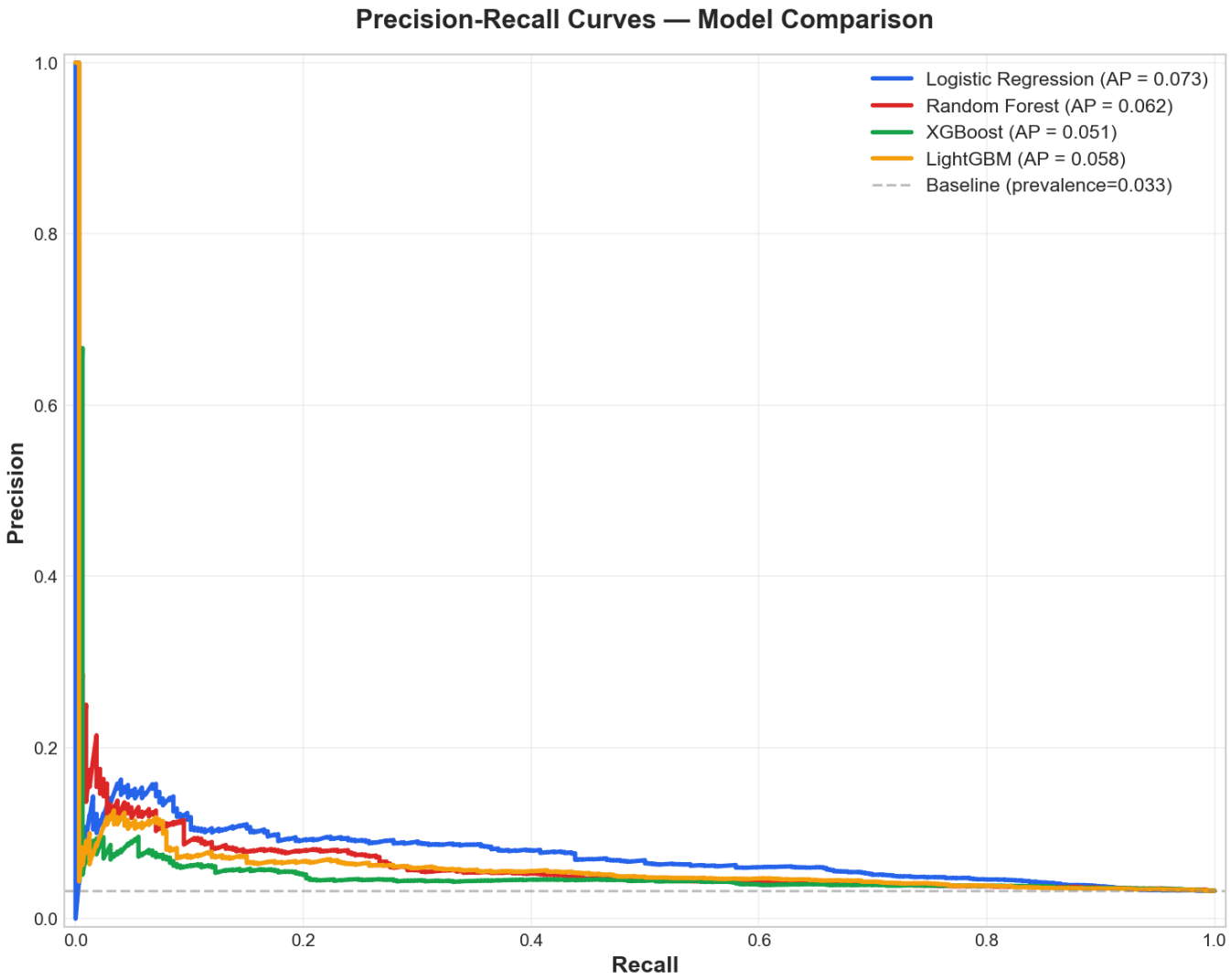
3. ROC & Precision-Recall Curves

ROC curves plot True Positive Rate vs False Positive Rate across all classification thresholds. AUC closer to 1.0 indicates better discrimination.

ROC Curves - All Models



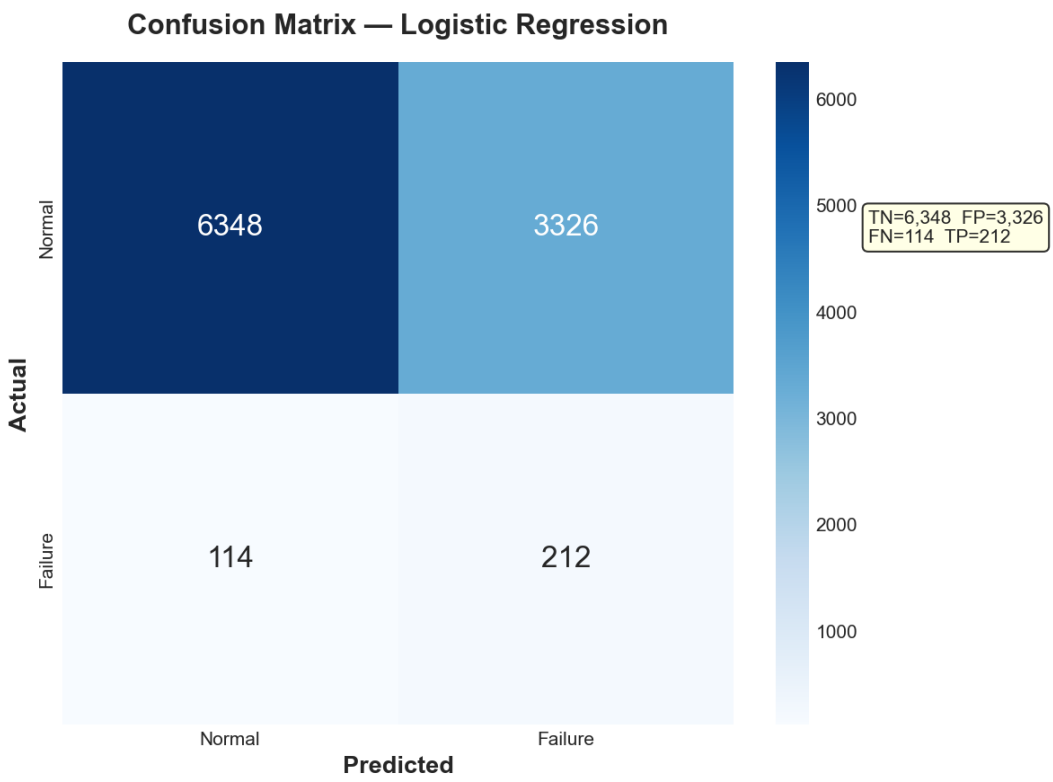
Precision-Recall Curves - All Models


Precision-Recall Curves — Model Comparison
Logistic Regression (AP = 0.073)
Random Forest (AP = 0.062)
XGBoost (AP = 0.051)
LightGBM (AP = 0.058)
Baseline (prevalence=0.033)
Precision
Recall
0.0
0.2
0.4
0.6
0.8
1.0

4. Confusion Matrices

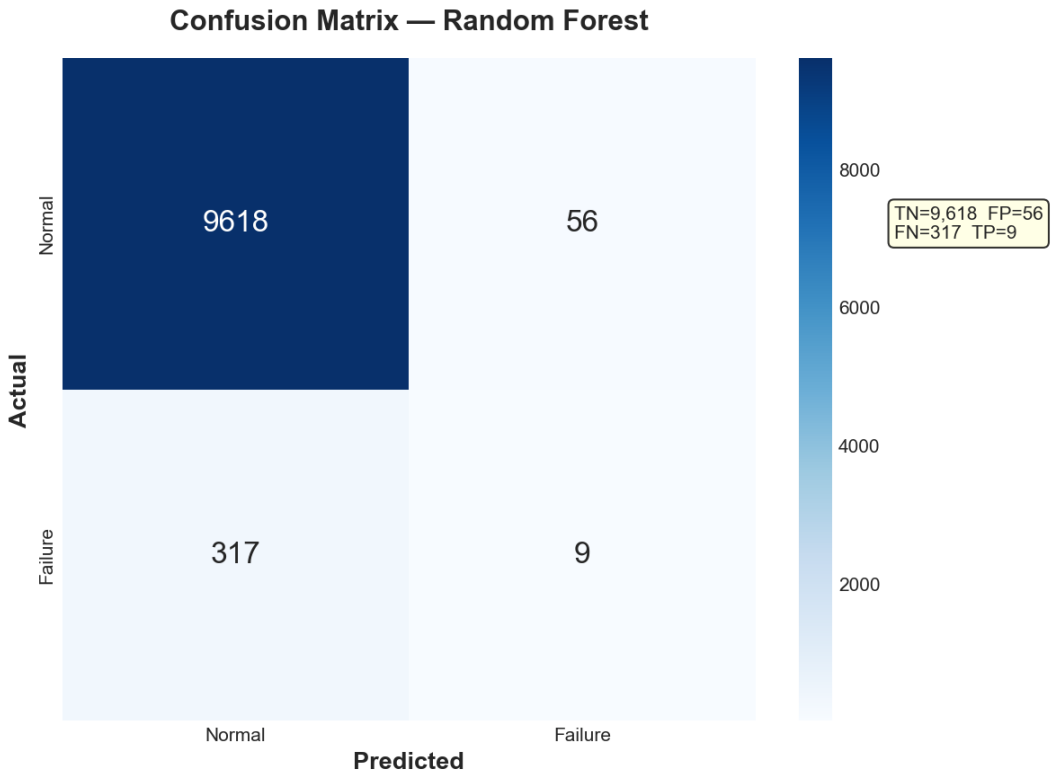
Confusion matrices show the breakdown of correct and incorrect predictions.

Confusion Matrix - Logistic Regression



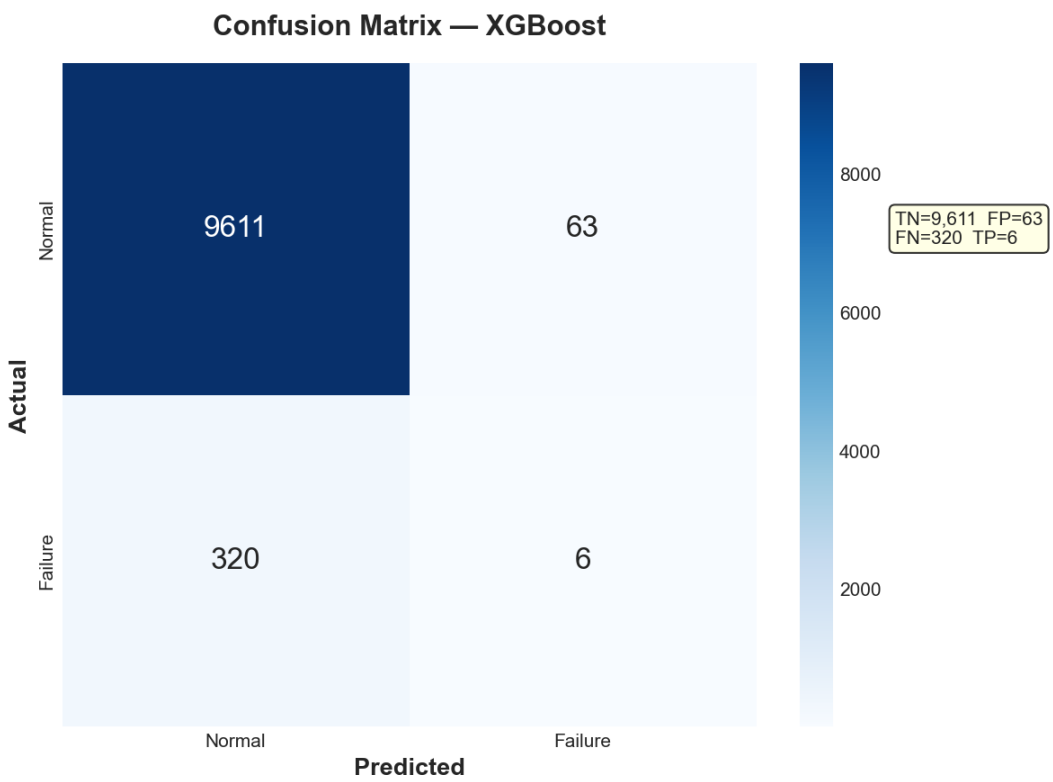
TN: 6,348 | FP: 3,326 | FN: 114 | TP: 212

Confusion Matrix - Random Forest



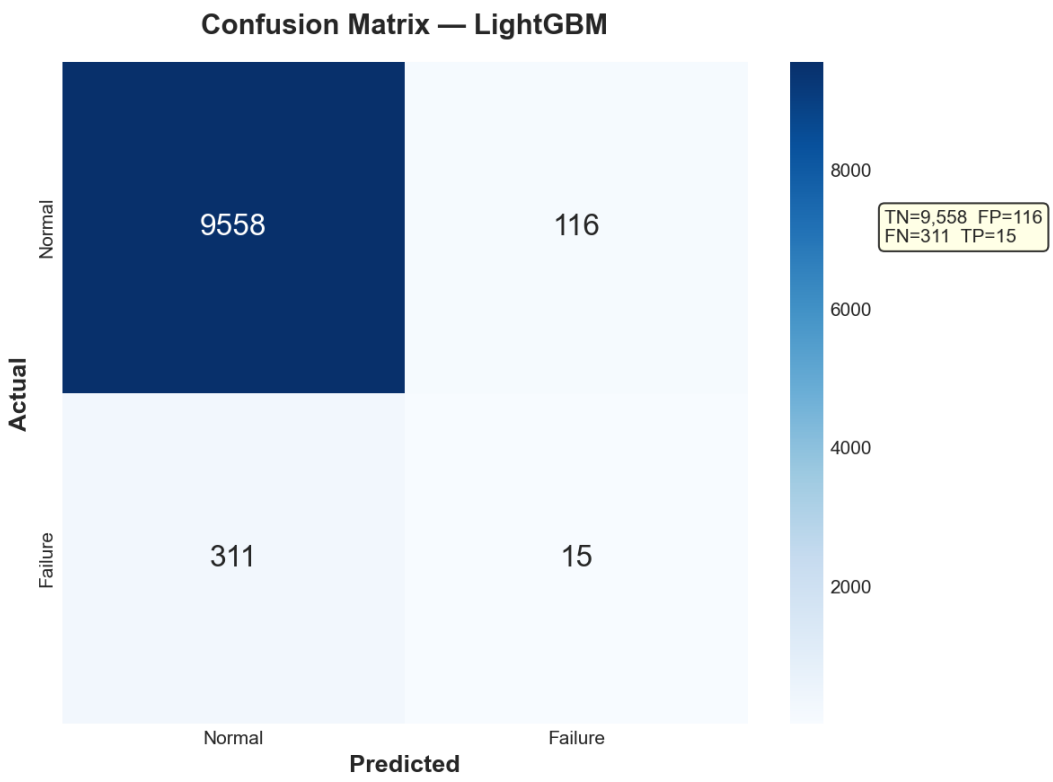
TN: 9,618 | FP: 56 | FN: 317 | TP: 9

Confusion Matrix - XGBoost



TN: 9,611 | FP: 63 | FN: 320 | TP: 6

Confusion Matrix - LightGBM

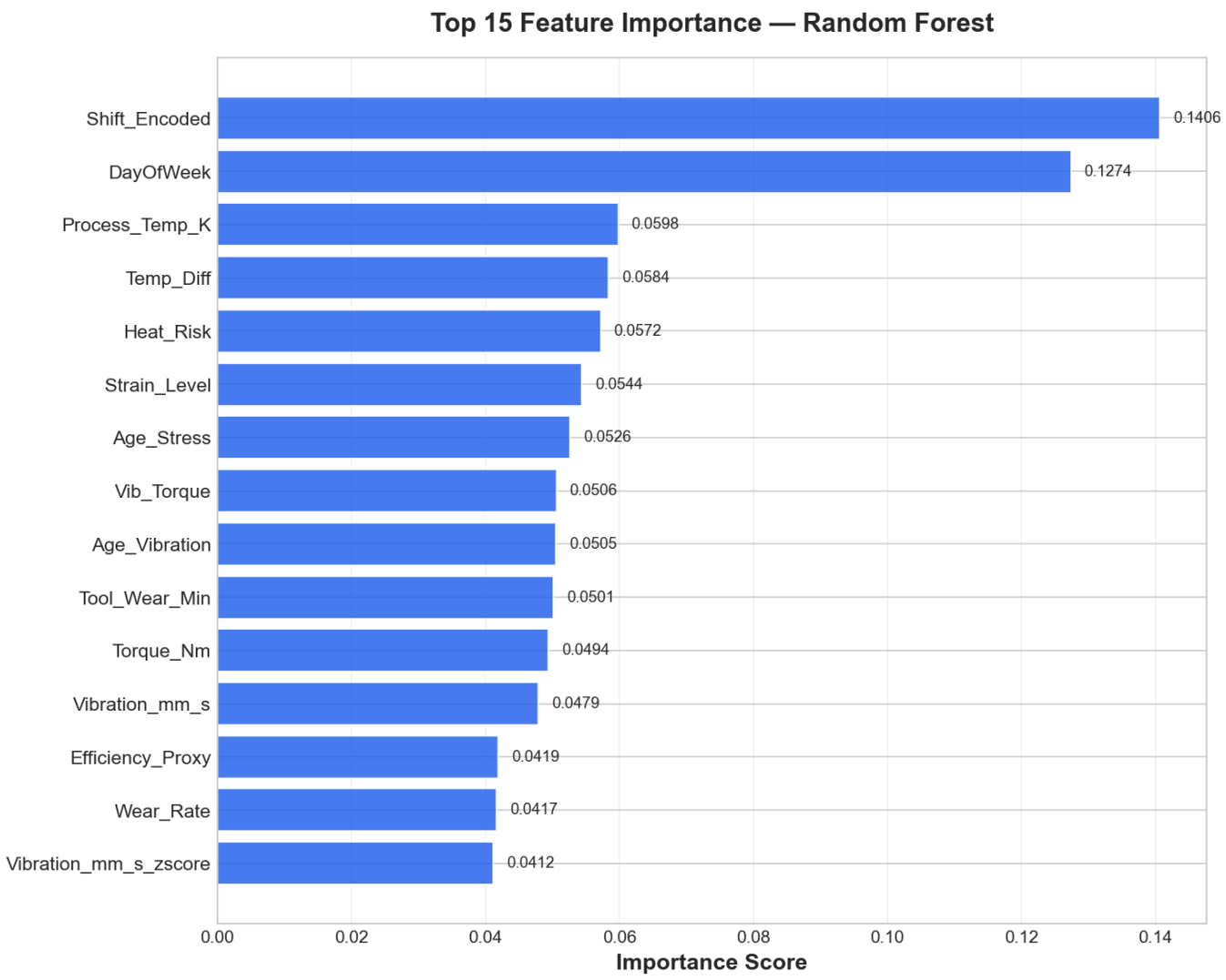


TN: 9,558 | FP: 116 | FN: 311 | TP: 15

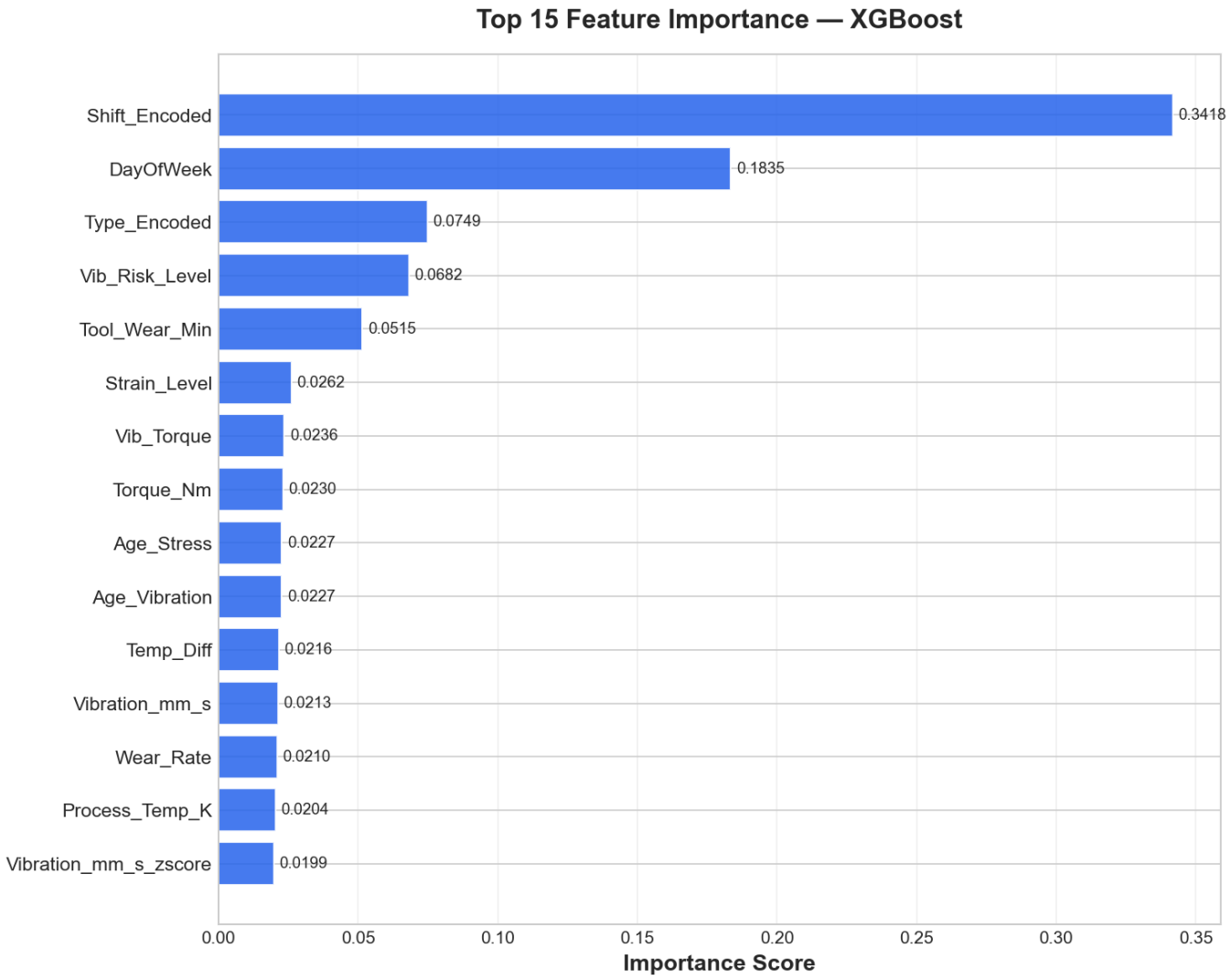
5. Feature Importance

Feature importance measures how much each feature contributes to predictions.

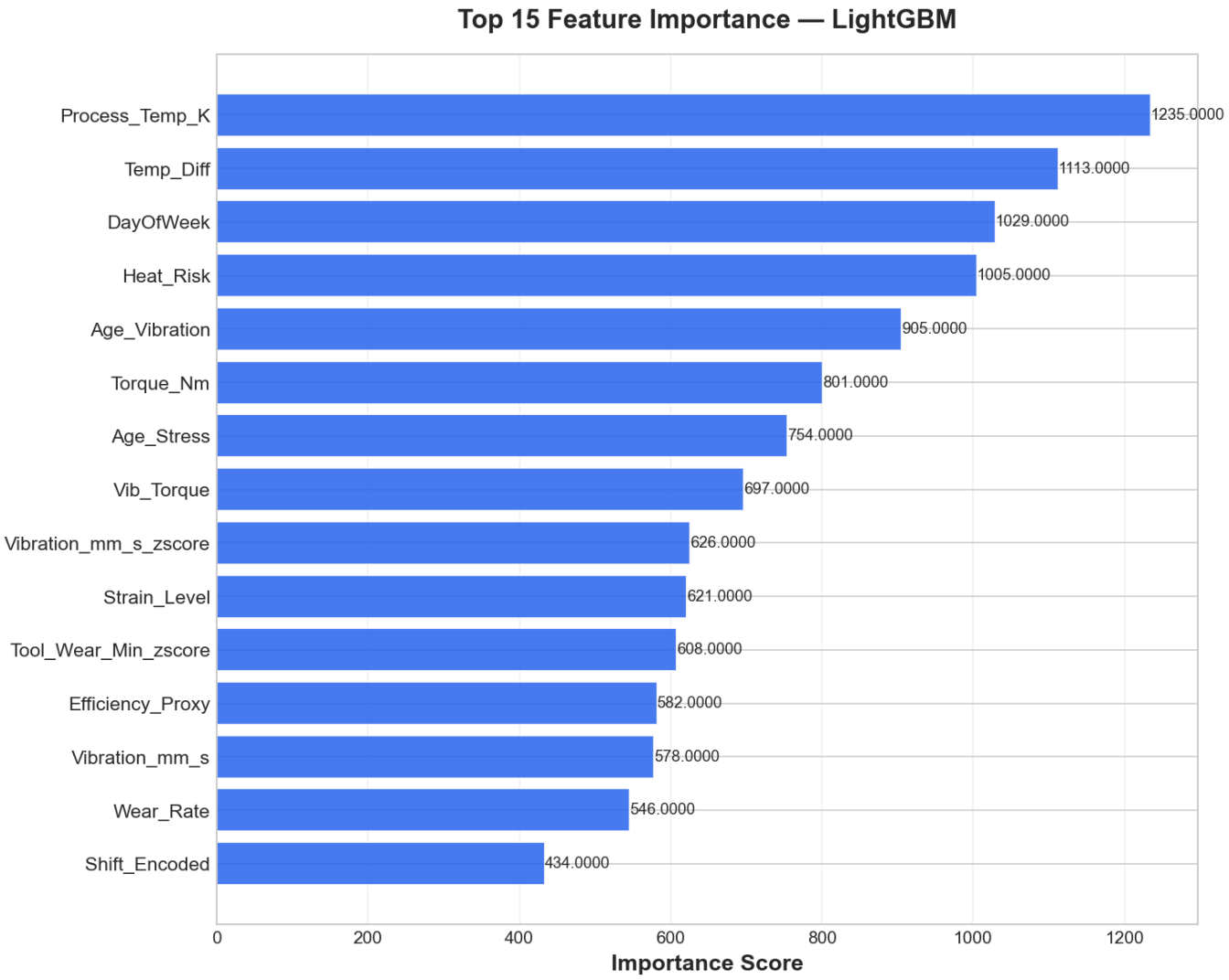
Feature Importance - Random Forest



Feature Importance - XGBoost



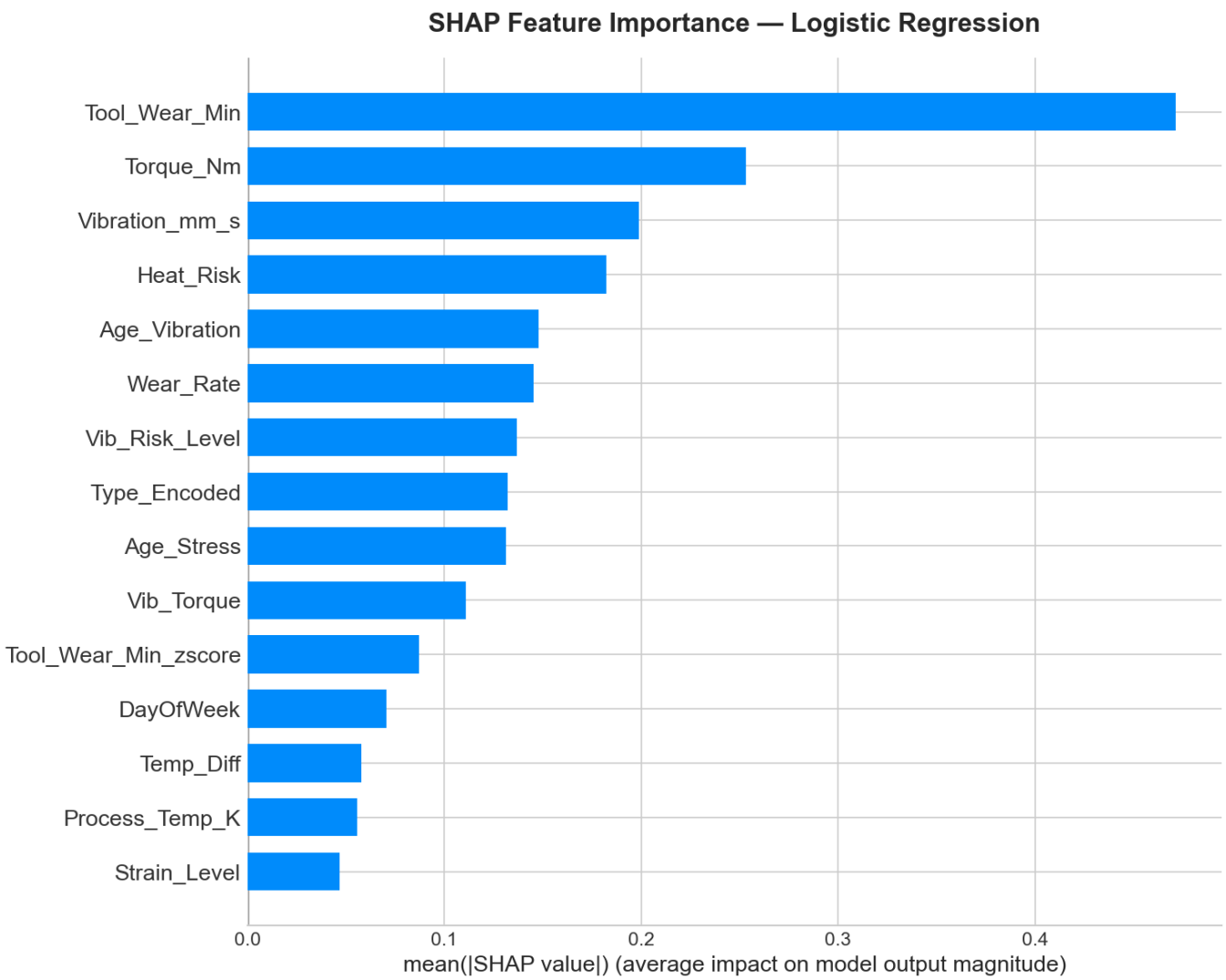
Feature Importance - LightGBM


Top 15 Feature Importance — LightGBM
Process_Temp_K
Temp_Diff
DayOfWeek
Heat_Risk
Age_Vibration
Torque_Nm
Age_Stress
Vib_Torque
Vibration_mm_s_zscore
Strain_Level
Tool_Wear_Min_zscore
Efficiency_Proxy
Vibration_mm_s
Wear_Rate
Shift_Encoded
1235.0000
1113.0000
1029.0000
1005.0000
905.0000
801.0000
754.0000
697.0000
626.0000
621.0000
608.0000
582.0000
578.0000
546.0000
434.0000
0
200
400
600
800
1000
1200
Importance Score

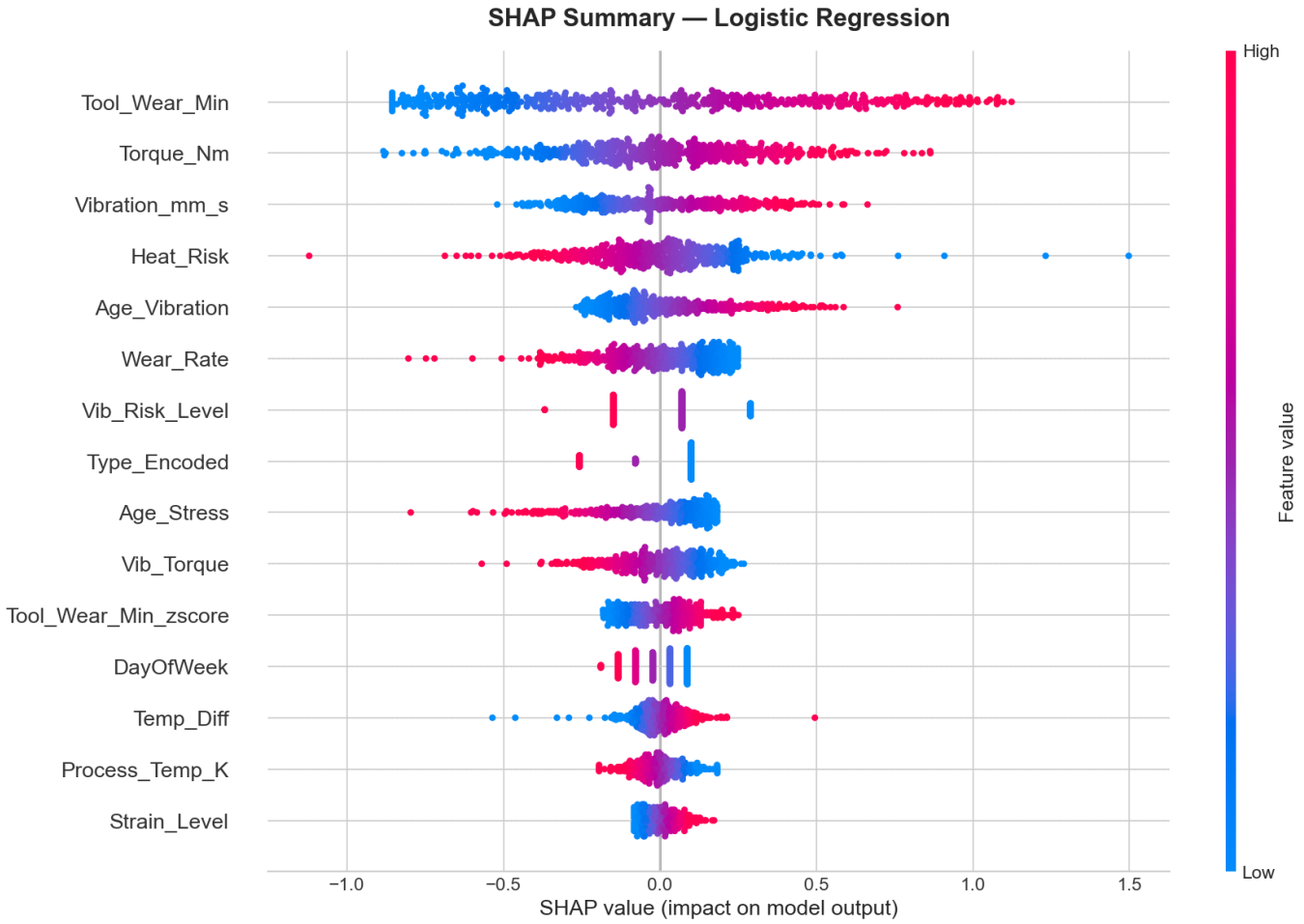
6. SHAP Explainability Analysis

SHAP provides mathematically grounded explanations for model predictions.

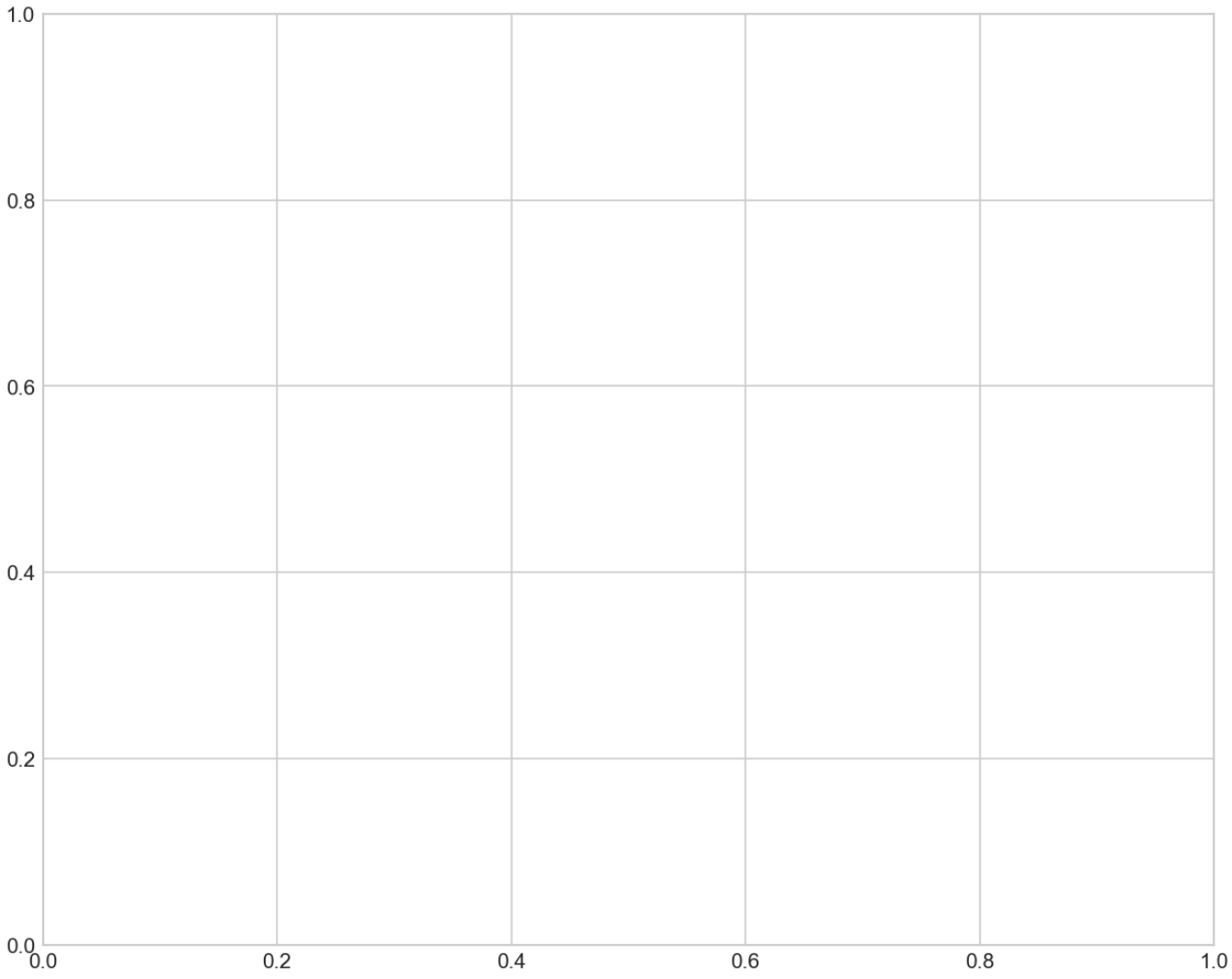
SHAP Global Importance - Logistic Regression



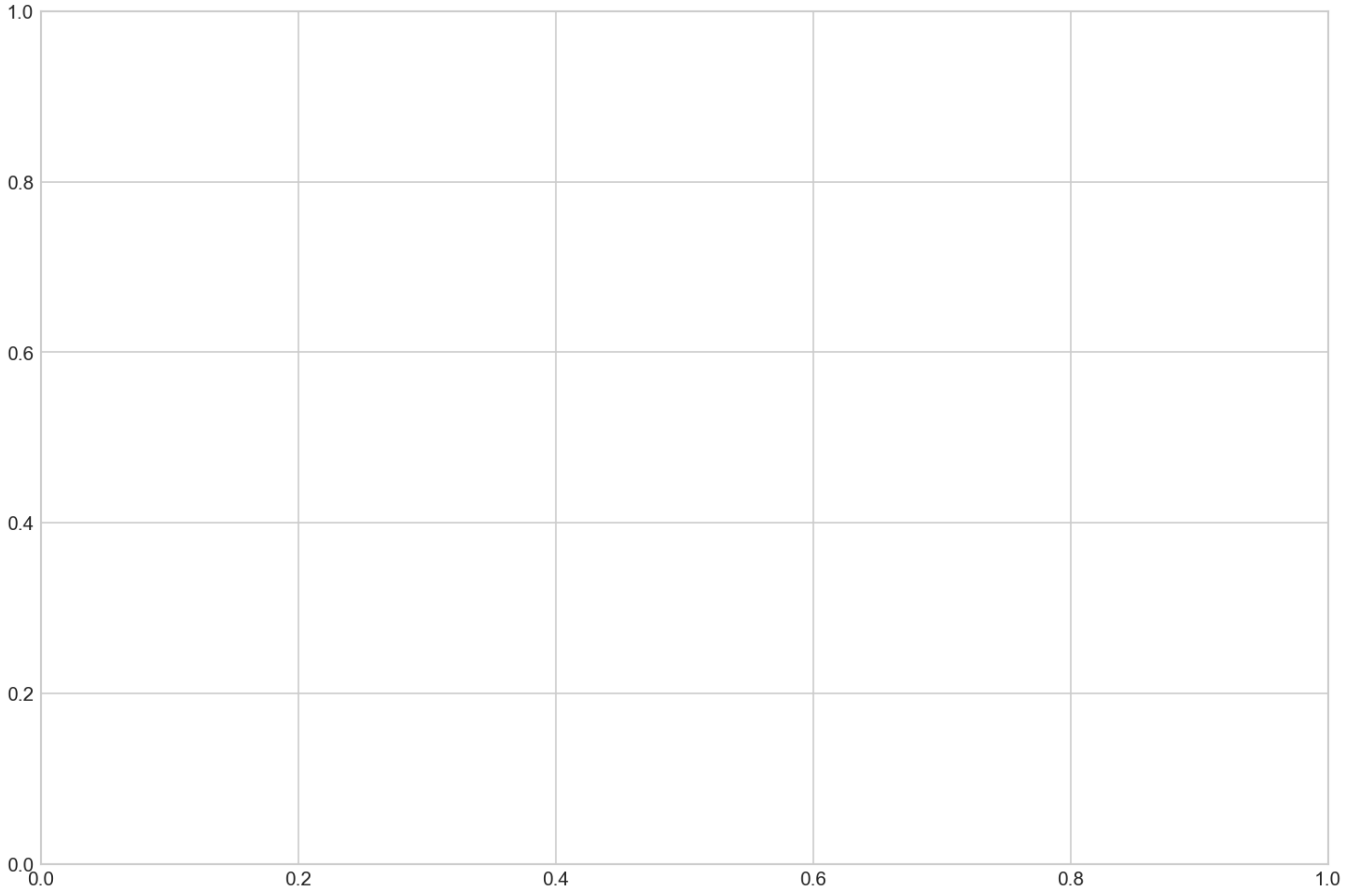
SHAP Beeswarm Plot - Logistic Regression



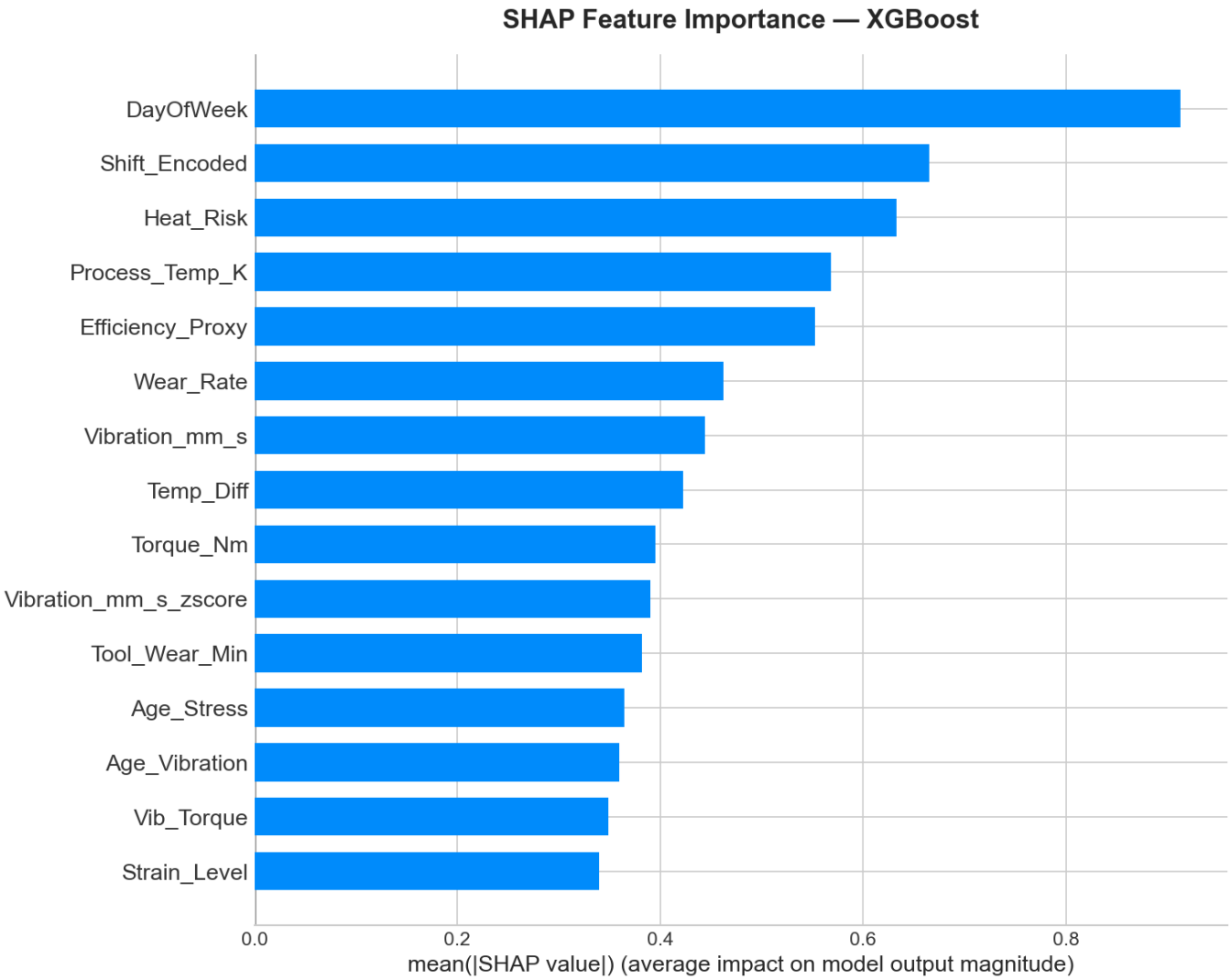
SHAP Global Importance - Random Forest



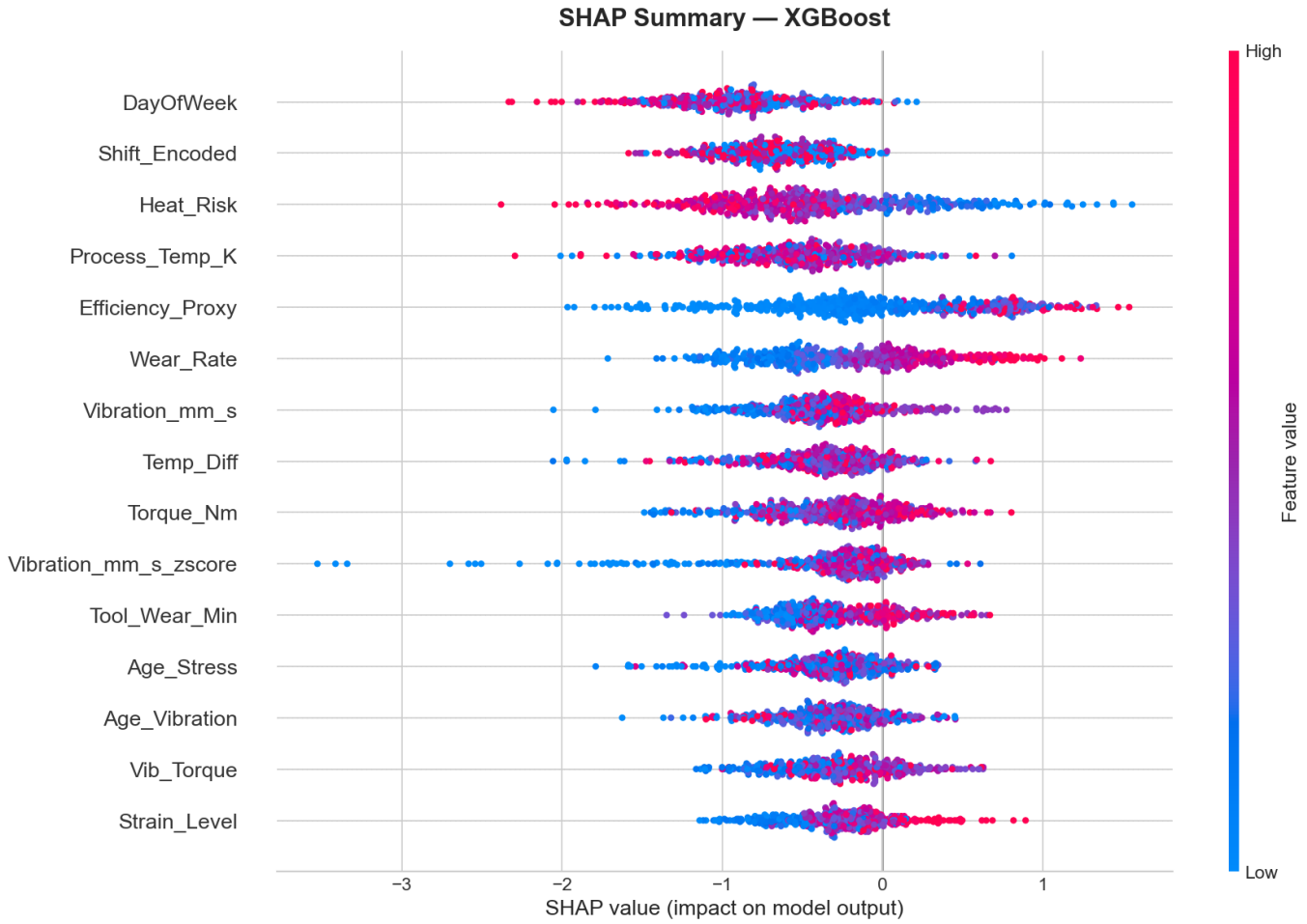
SHAP Beeswarm Plot - Random Forest



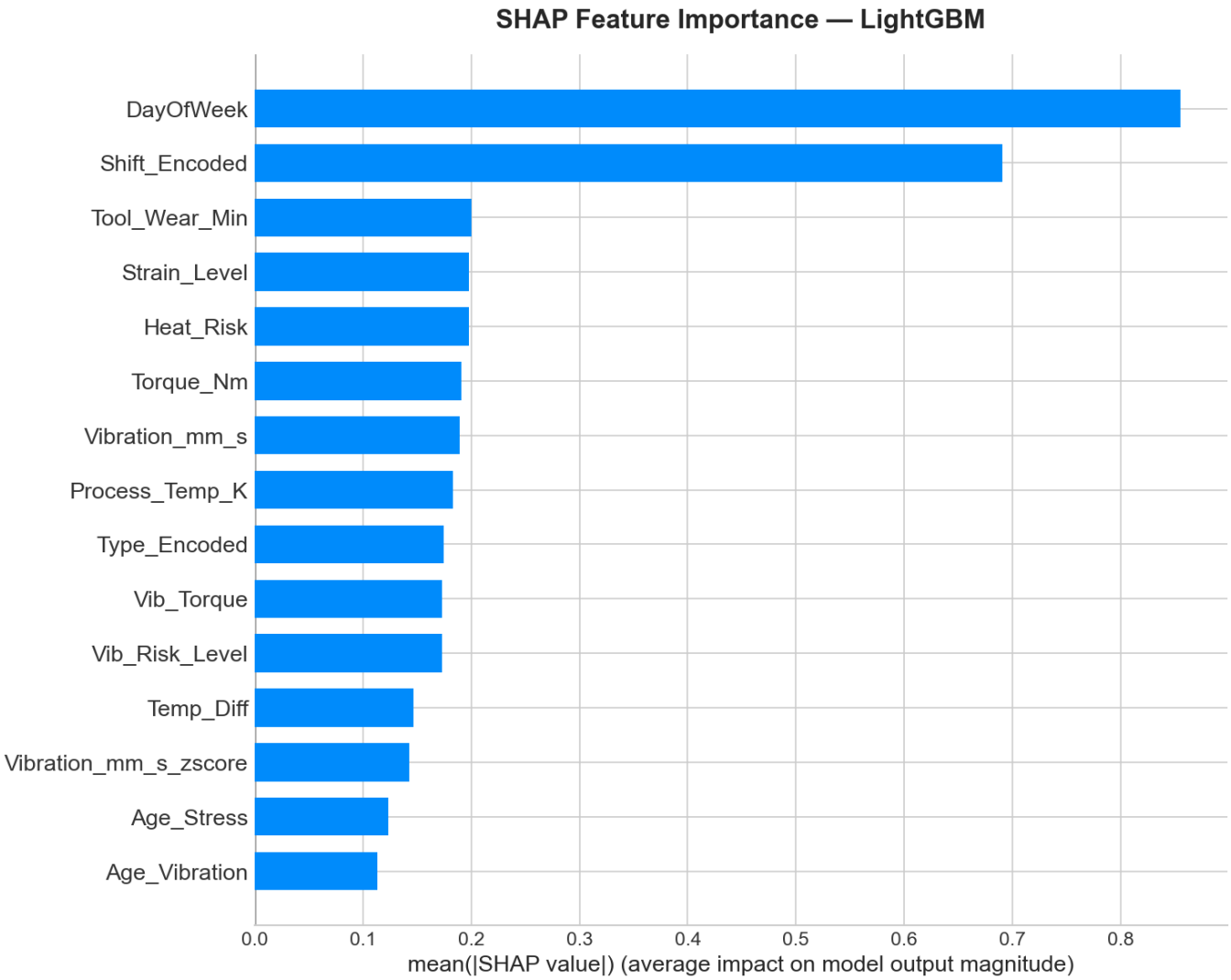
SHAP Global Importance - XGBoost



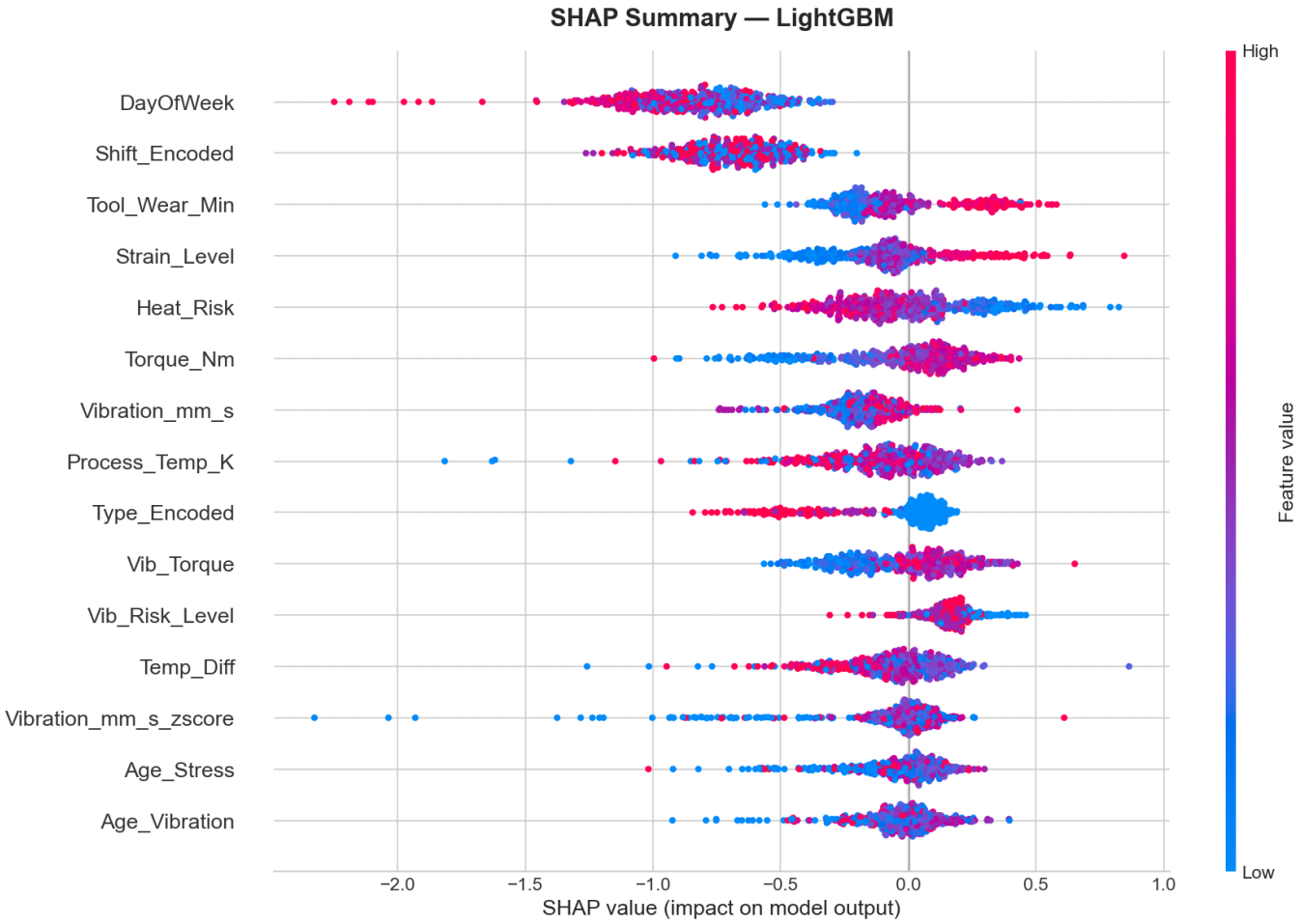
SHAP Beeswarm Plot - XGBoost



SHAP Global Importance - LightGBM



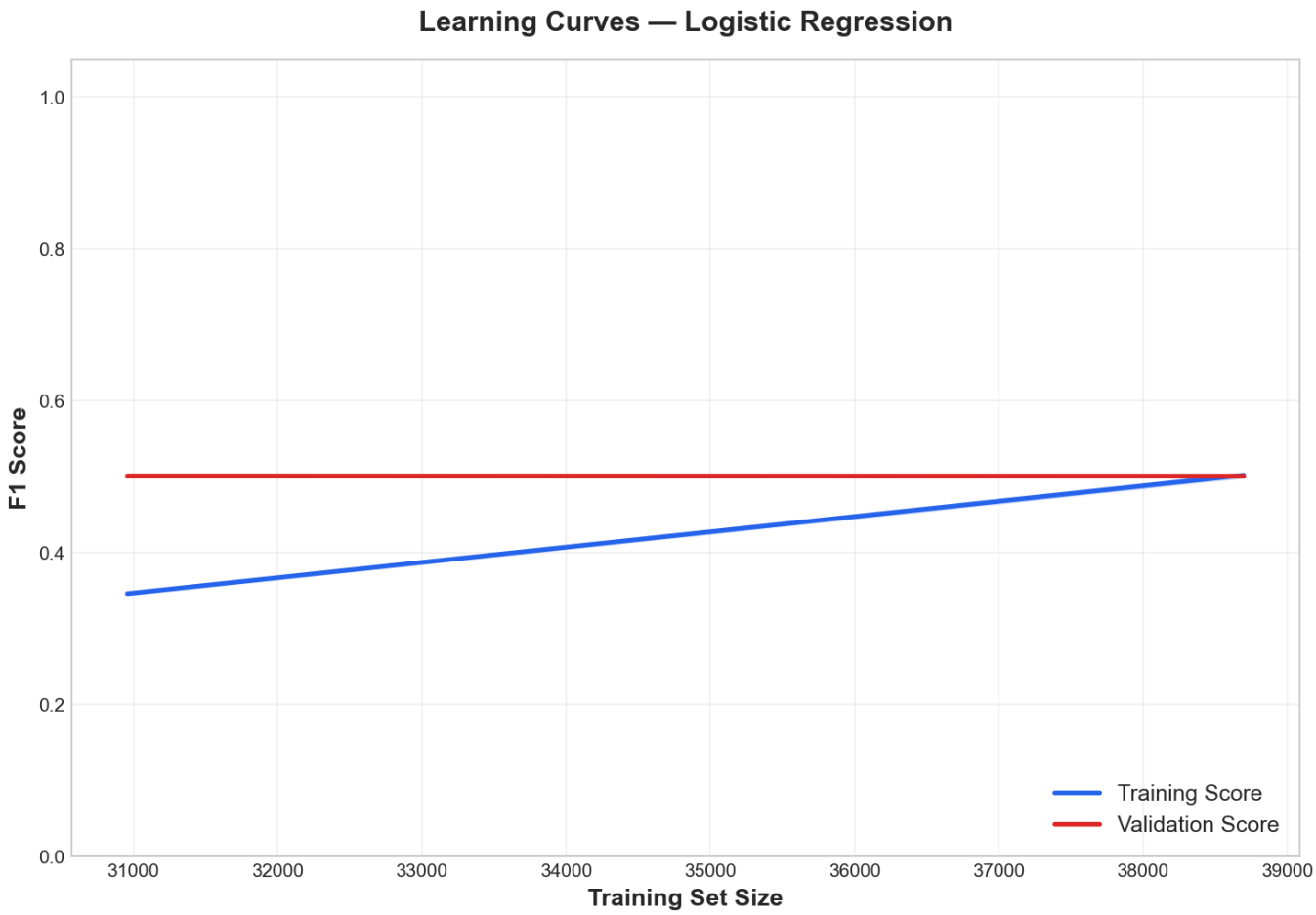
SHAP Beeswarm Plot - LightGBM


SHAP Summary — LightGBM
DayOfWeek
Shift_Encoded
Tool_Wear_Min
Strain_Level
Heat_Risk
Torque_Nm
Vibration_mm_s
Process_Temp_K
Type_Encoded
Vib_Torque
Vib_Risk_Level
Temp_Diff
Vibration_mm_s_zscore
Age_Stress
Age_Vibration
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
SHAP value (impact on model output)
High
Feature value
Low

7. Learning Curves

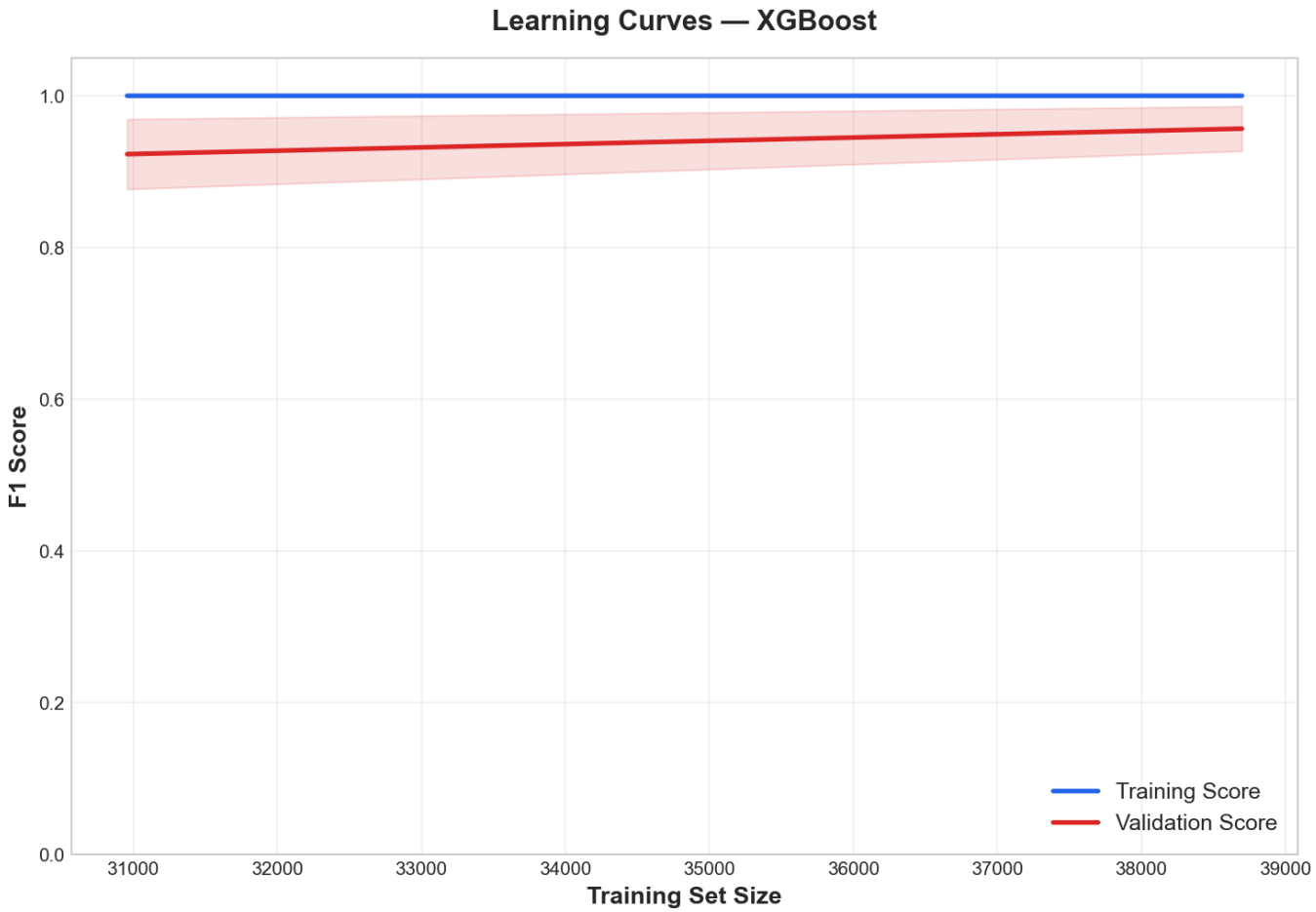
Learning curves show how model performance changes as training data increases.

Learning Curve - Logistic Regression

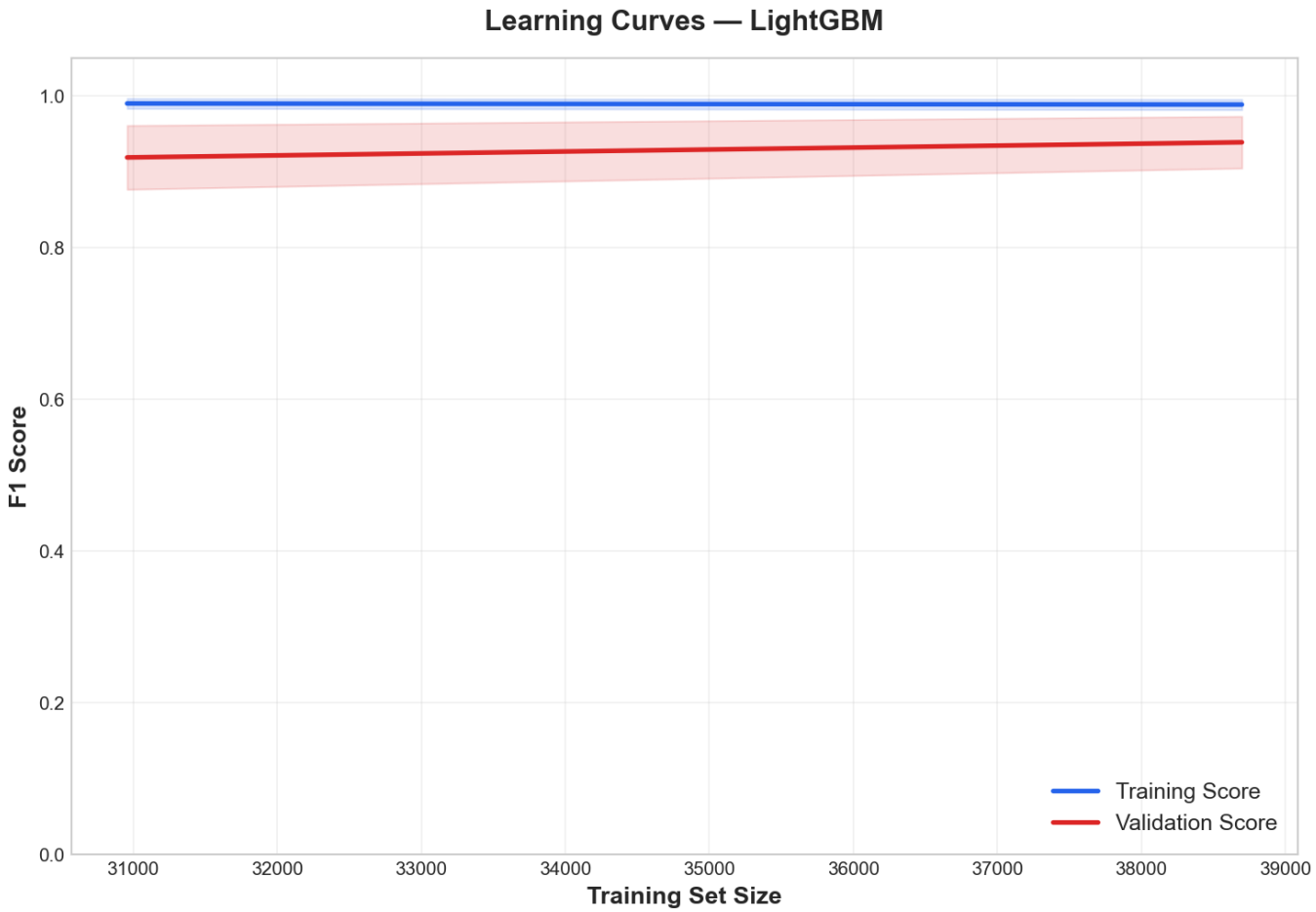


Learning Curve - Random Forest

Learning Curve - XGBoost



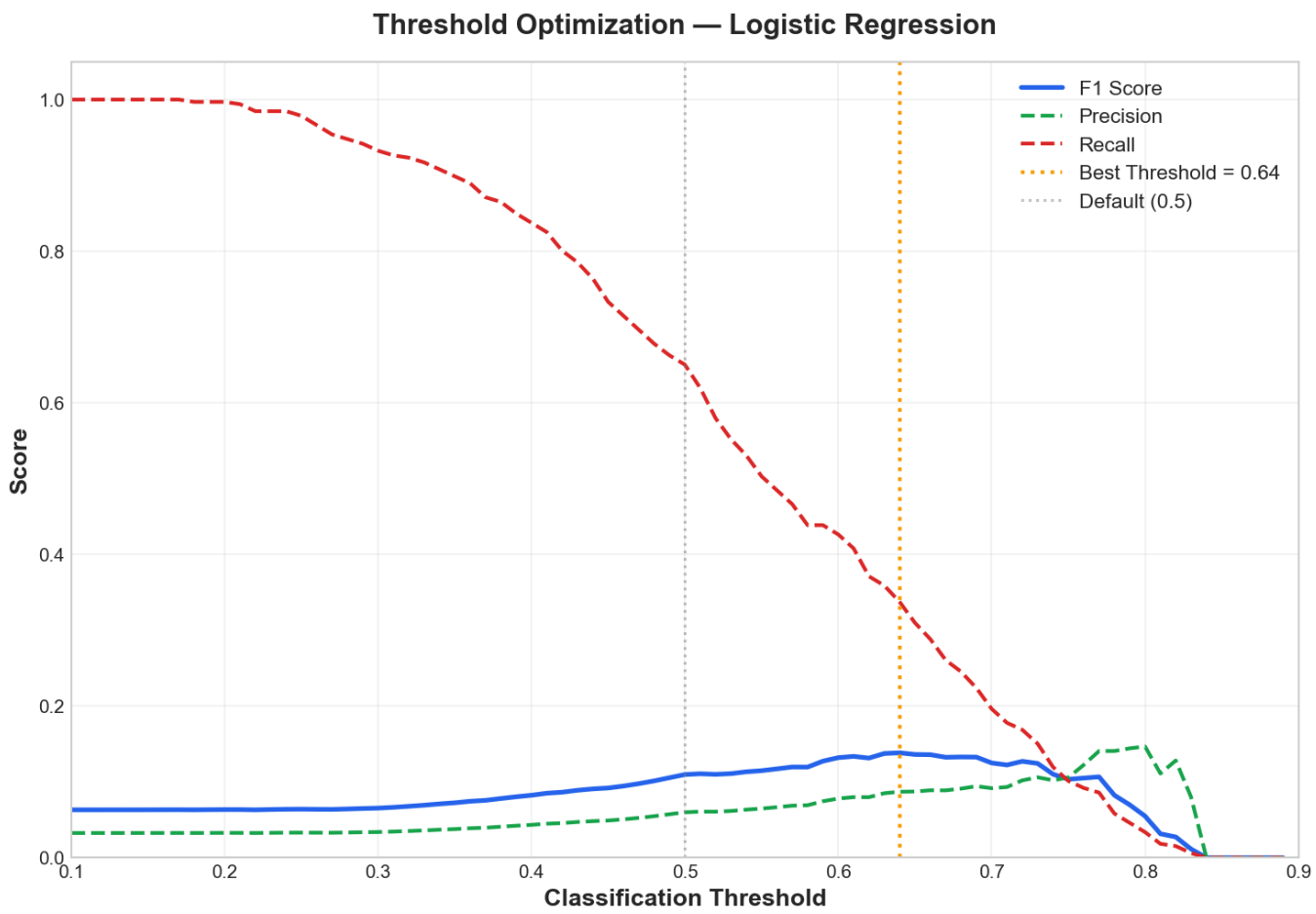
Learning Curve - LightGBM


Learning Curves — LightGBM
F1 Score
1.0
0.8
0.6
0.4
0.2
0.0
31000
32000
33000
34000
35000
36000
37000
38000
39000
Training Set Size
Training Score
Validation Score

8. Threshold Optimization

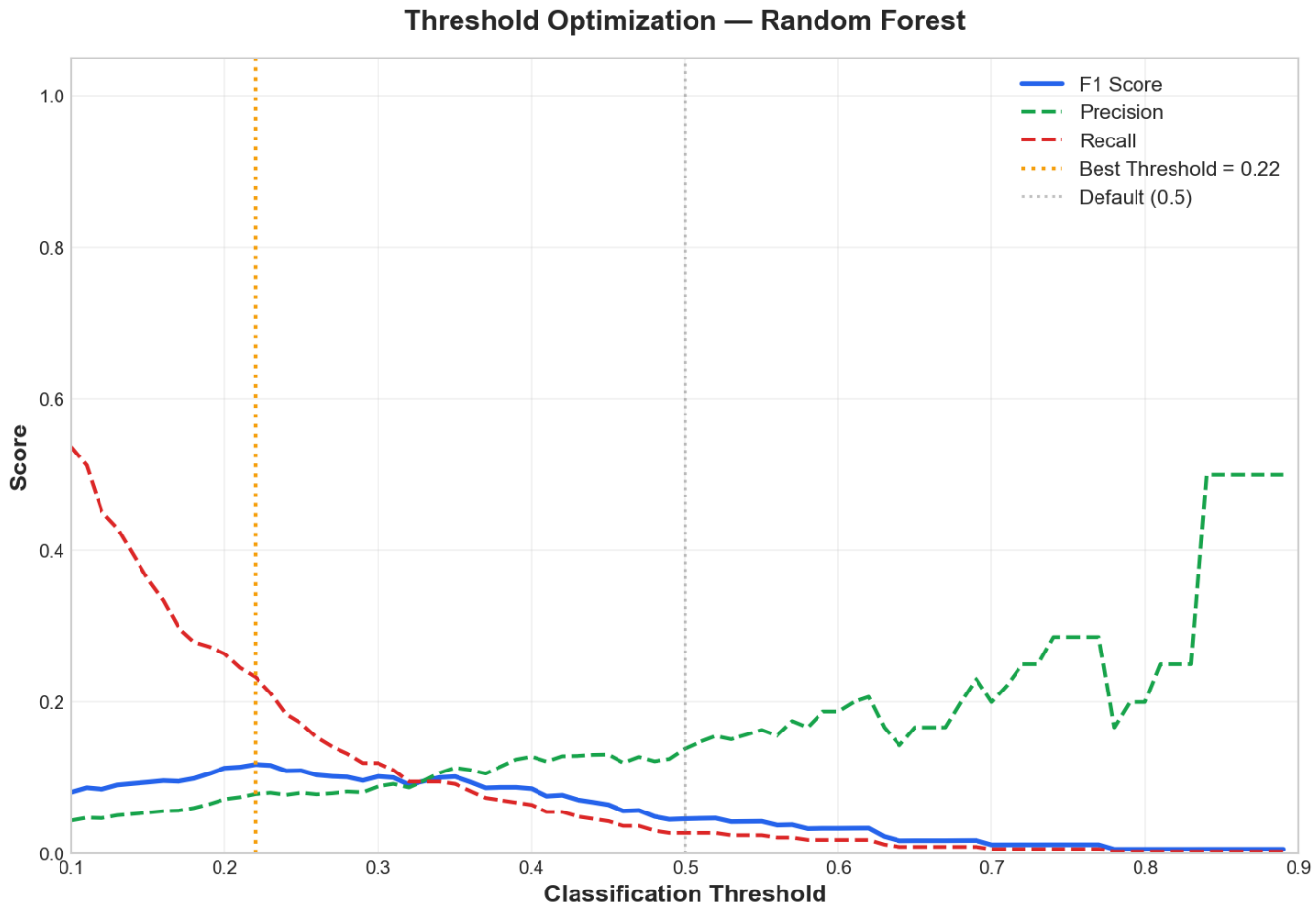
The default threshold of 0.5 is often suboptimal for imbalanced datasets.

Threshold Optimization - Logistic Regression



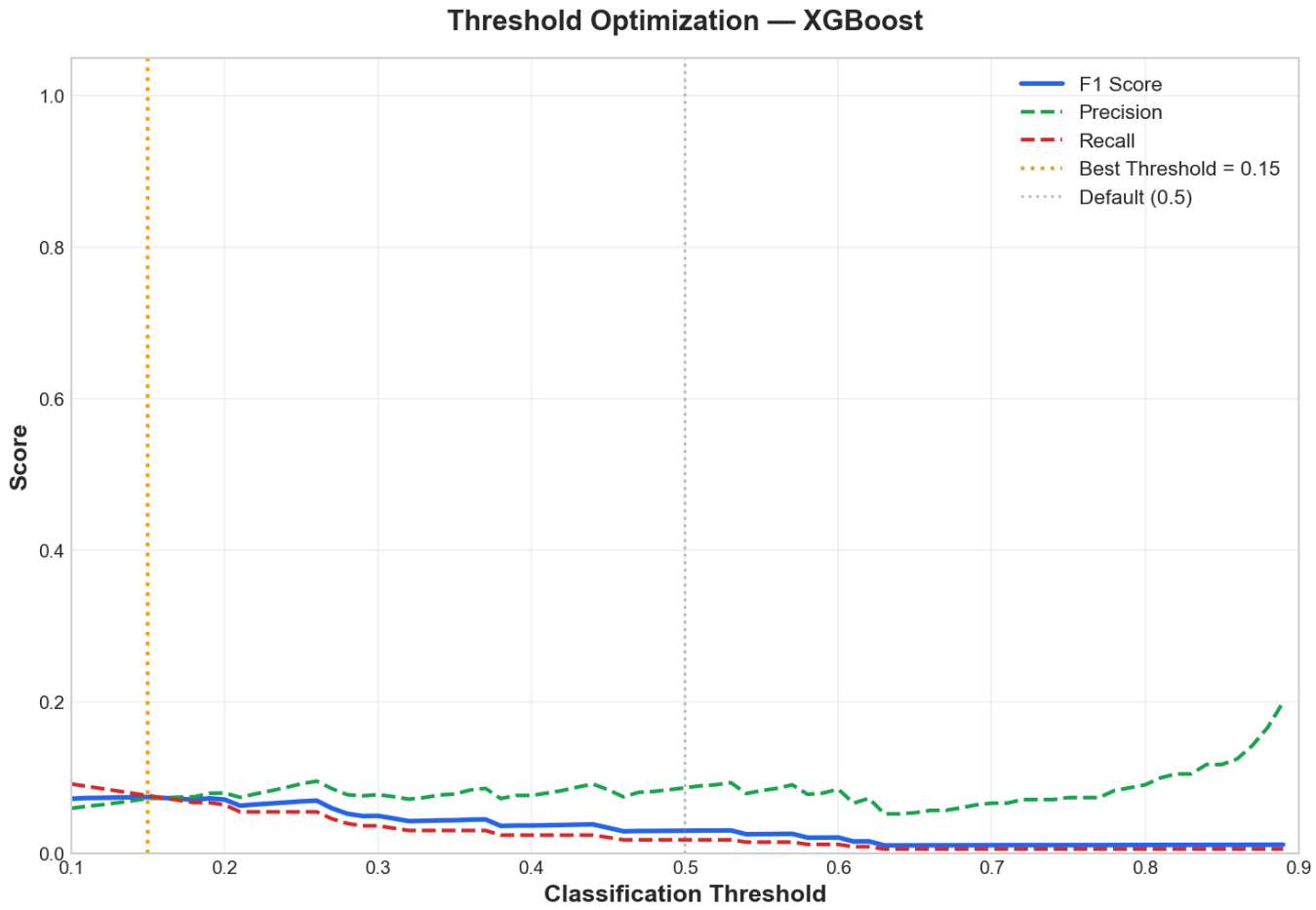
Optimal Threshold	0.64
Best F1	0.1384
Default F1 (0.5)	0.1097
Improvement	+0.0286

Threshold Optimization - Random Forest



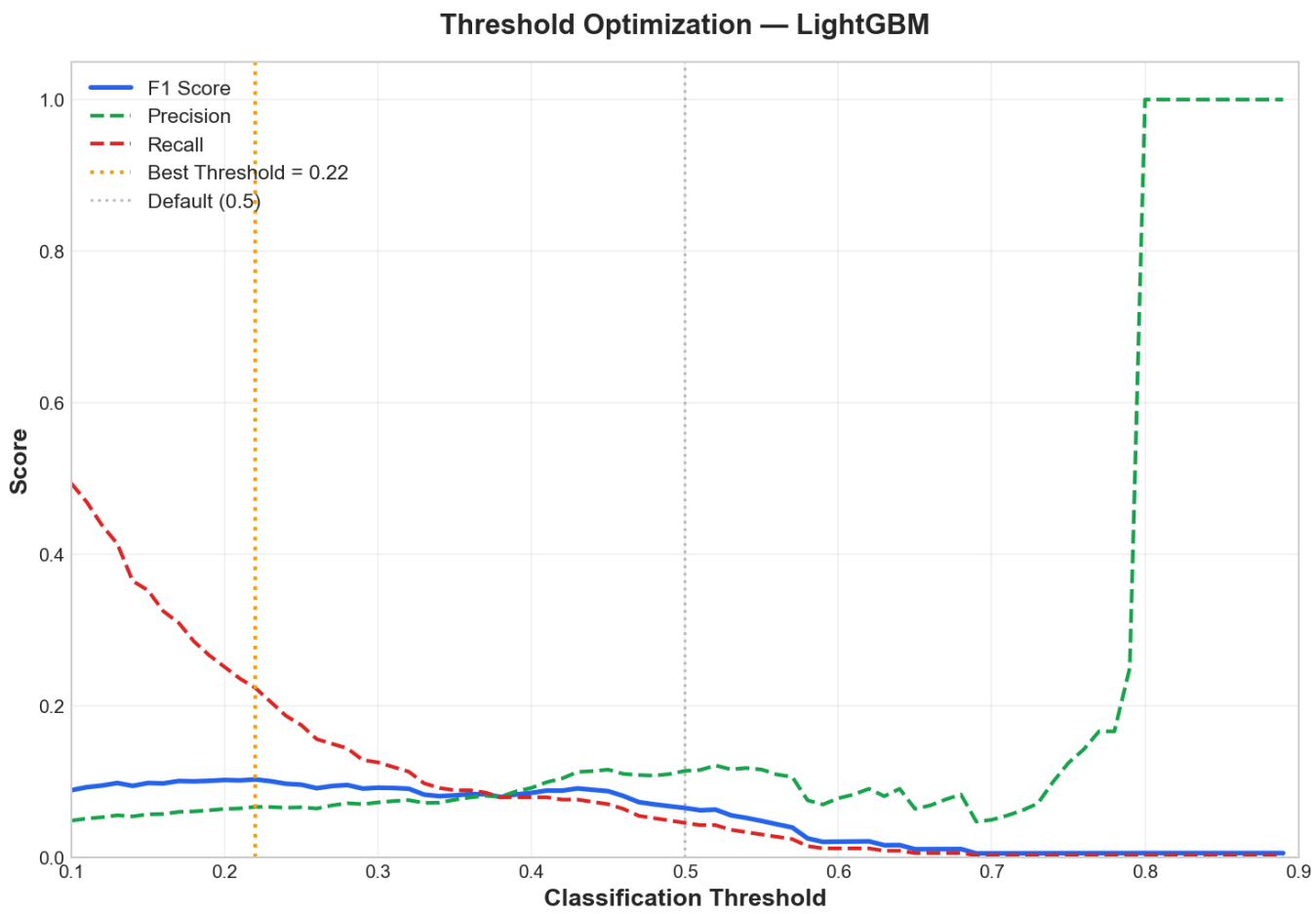
Optimal Threshold	0.22
Best F1	0.1178
Default F1 (0.5)	0.0460
Improvement	+0.0718

Threshold Optimization - XGBoost



Optimal Threshold	0.15
Best F1	0.0747
Default F1 (0.5)	0.0304
Improvement	+0.0444

Threshold Optimization - LightGBM



Optimal Threshold	0.22
Best F1	0.1033
Default F1 (0.5)	0.0656
Improvement	+0.0376

9. Best Hyperparameters

Hyperparameters were optimized using RandomizedSearchCV.

Logistic Regression

Description: Linear baseline - fast, interpretable

Parameter	Value
solver	liblinear
penalty	l2
C	100

Training Time 19.7s

Random Forest

Description: Ensemble of decision trees - robust, handles non-linearity

Parameter	Value
n_estimators	200
min_samples_split	5
min_samples_leaf	1
max_features	log2
max_depth	None

Training Time 1881.4s

XGBoost

Description: Gradient boosting - high accuracy, handles imbalance

Parameter	Value
subsample	0.9
n_estimators	300
min_child_weight	1
max_depth	10
learning_rate	0.2
gamma	0

colsample_bytree	1.0

Training Time 158.7s

LightGBM

Description: Fast gradient boosting - efficient, great with large data

Parameter	Value
subsample	0.8
num_leaves	63
n_estimators	200
min_child_samples	10
max_depth	-1
learning_rate	0.1
colsample_bytree	0.9

Training Time 149.9s

10. Pipeline Metadata

Technical details about the ETL pipeline and data processing steps.

Pipeline Timestamp	2026-02-24T16:18:07.695456
Original Data Shape	[50000, 19]
Steps Completed	9
Missing Values Filled	6950
Outliers Treated	2112

Pipeline Steps Executed

1. validation
2. imputation
3. outlier_treatment
4. feature_engineering
5. build_matrix
6. split
7. scaling
8. feature_selection
9. imbalance_smote

11. Test Set Prediction Analysis

The test set contains 10,000 samples. Actual failures: 326 (3.3%). Predicted failures: 3,538 (35.4%).

Prediction Probability Statistics

Mean Probability	0.4626
Median Probability	0.4457
Std Deviation	0.1388
Min Probability	0.0714
Max Probability	0.8794

12. Selected Features

The final model uses 18 features after selection.

#	Feature Name
1	Vib_Torque
2	Strain_Level
3	Tool_Wear_Min
4	Vibration_mm_s
5	Tool_Wear_Min_zscore
6	Vib_Risk_Level
7	Age_Vibration
8	Wear_Rate
9	Age_Stress
10	Efficiency_Proxy
11	Torque_Nm
12	Vibration_mm_s_zscore
13	Temp_Diff
14	Type_Encoded
15	Heat_Risk
16	Shift_Encoded
17	DayOfWeek
18	Process_Temp_K

End of Report

PredictiveGuard(TM) - Industrial Predictive Maintenance System

Generated: March 10, 2026 at 11:45:59

Built with Scikit-Learn | XGBoost | LightGBM | SHAP | Streamlit